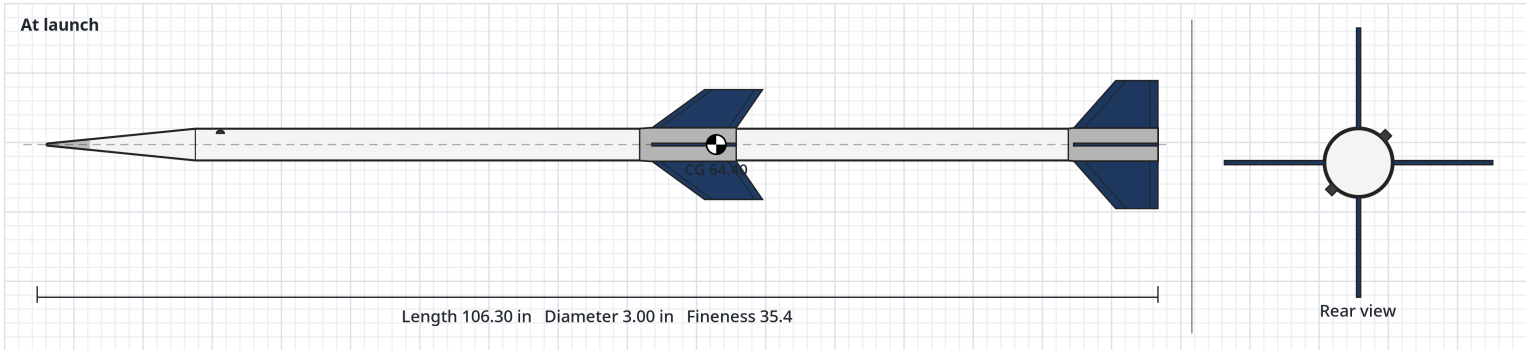
PANDORA - EAYF_CONFIG_1 - Results



PANDORA - EAYF_CONFIG_1
Nose radius for heating: 0.1 in

	nominal	mean	std	-2 sigma	+2 sigma	95% low	95% high	lowest	highest	
Rail exit speed	114.08	114.12	1.2544	111.61	116.63	111.66	116.58	111.36	116.51	ft/s
Rail exit AoA	2.2067	3.306	2.62	-1.9339	8.5459	-1.8343	8.4463	2.0041	2.8559	deg
Forecast gust	13.123	13.298	6.2137	0.87046	25.725	1.1067	25.489	12.139	15.092	ft/s
Rail exit AoA in gust	6.5623	6.6235	3.0442	0.53513	12.712	0.65088	12.596	6.2263	7.382	deg
Apogee	1.1825e+05	1.1578e+05	7089.2	1.0161e+05	1.2996e+05	1.0187e+05	1.2969e+05	94149	1.3987e+05	ft
Time to apogee	88.959	87.833	3.138	81.557	94.109	81.677	93.99	77.584	99.131	s
Max speed	4356.3	4356.5	92.31	4171.9	4541.1	4175.4	4537.6	4216.4	4471.8	ft/s
Max acceleration	73.909	73.888	1.8617	70.164	77.611	70.235	77.54	71.343	76.431	g
Max roll rate	0.0040759	6.3925	4.2505	-2.1085	14.893	-1.9468	14.732	13.271	12.315	rev/s
Max angle of attack	0.14378	1.248	0.48648	0.275	2.2209	0.2935	2.2024	2.4969	0.90933	deg
Max AoA after burnout	0.013665	0.41536	0.40141	-0.38746	1.2182	-0.3722	1.2029	0.92013	0.27184	deg
Roll-pitch crossings	0	0.70021	0.79379	-0.88738	2.2878	-0.85719	2.2576	1	1	
Resonance time	nan	23.166	9.609	3.9483	42.385	4.2985	42.034	14.287	16.465	s
Resonance altitude	nan	41400	23732	-6064.8	88864	-5200	87999	14924	18033	ft
AoA rise at resonance	nan	0.66554	0.6082	-0.55087	1.8819	-0.52871	1.8598	0.50738	0.11354	deg
Roll lock-in held	0	0	0	0	0	0	0	0	0	s
Lock-in starts	nan	(no run)	s							
Max Mach	4.1205	4.1161	0.091006	3.9341	4.2981	3.9375	4.2946	3.9603	4.2551	
Max dynamic pressure	11231	11252	440.21	10371	12132	10388	12115	11008	11173	psf
Peak heat flux	554.61	555.05	33.781	487.49	622.61	488.77	621.33	515.03	580.4	BTU/ft²/s
Heat load	2975.9	2977.6	182.44	2612.7	3342.5	2619.7	3335.6	2656	3318.2	BTU/ft²
Tilt at ignition	3.984	6.9169	3.3973	0.12229	13.712	0.25147	13.582	14.854	4.4059	deg
Altitude at ignition	12050	12006	429.77	11147	12866	11163	12849	10965	13630	ft
Flight time	369.81	367.19	8.0224	351.14	383.23	351.45	382.93	342.95	387.02	s
Landing east	-1.3636	-1.1423	5.3474	-11.837	9.5525	-11.634	9.3492	-2.5941	1.8238	mi
Landing north	-0.99616	-0.45797	5.2623	-10.983	10.067	-10.783	9.8666	-10.705	-5.9816	mi
Landing distance	1.6887	6.9225	3.128	0.66645	13.179	0.78539	13.06	11.015	6.2535	mi
Impact speed	40.683	41.32	1.5811	38.158	44.482	38.218	44.422	40.689	41.434	ft/s
Launch mass	38.822	lb, CG 64.40 in from the nose tip								
Min static margin	3.1295	cal (while loaded)								

Angles of attack are while loaded, launch to apogee: dynamic pressure above 5% of its peak.
Failure conditions: lock-in = true; resonance = true; lockout = true; Landing distance > 8. The nominal flight passes; 234 of 512 Monte Carlo runs fail (45.7%
Sustainer ignited: yes
  Monte Carlo: 487 of 512 ignited, 25 tilt lockout
Monte Carlo runs: 512
  -2 sigma and +2 sigma are the mean less and plus two standard deviations.
  95% low and high: the prediction interval, where 95% of flights fall: mean -/+ z s sqrt(1 + 1/N),
  z = 1.960, N = 487 runs, assuming a normal spread.

PANDORA - EAYF_CONFIG_1 - Weather and Monte Carlo inputs

Weather

Open-Meteo forecast at 35.3471, -117.8082, 17:00 UTC

day	surface wind	gust at the launch hour	runs	
2026-09-16	9.2 ft/s from 202 deg	13.1 ft/s from 274 deg at 17:00	37	nominal flight
2026-09-15	4.9 ft/s from 187 deg	9.2 ft/s from 284 deg at 17:00	26	
2026-09-17	12.4 ft/s from 199 deg	13.1 ft/s from 223 deg at 17:00	30	
2026-09-14	2.3 ft/s from 77 deg	12.8 ft/s from 241 deg at 17:00	32	
2026-09-18	5.1 ft/s from 212 deg	5.9 ft/s from 233 deg at 17:00	34	
2026-09-13	18.9 ft/s from 240 deg	30.8 ft/s from 237 deg at 17:00	32	
2026-09-19	5.2 ft/s from 22 deg	19.4 ft/s from 244 deg at 17:00	41	
2026-09-12	7.7 ft/s from 218 deg	12.1 ft/s from 231 deg at 17:00	32	
2026-09-20	6.0 ft/s from 28 deg	17.1 ft/s from 263 deg at 17:00	35	
2026-09-11	15.7 ft/s from 273 deg	19.0 ft/s from 211 deg at 17:00	31	
2026-09-21	0.0 ft/s from 180 deg	3.0 ft/s from 312 deg at 17:00	41	
2026-09-10	3.1 ft/s from 315 deg	8.2 ft/s from 247 deg at 17:00	38	
2026-09-22	5.3 ft/s from 9 deg	11.2 ft/s from 240 deg at 17:00	36	
2026-09-09	5.0 ft/s from 19 deg	14.1 ft/s from 244 deg at 17:00	37	
2026-09-23	5.4 ft/s from 25 deg	15.1 ft/s from 272 deg at 17:00	30	

Each day is a whole forecast wind profile; the gust is the 10 m forecast for the launch hour, used only for the rail exit AoA in gust.

Monte Carlo inputs: these 18 change from run to run, everything else is fixed

	nominal	sigma	lowest	mean	highest	(as drawn)
Booster impulse	100	2	94.22	100.1	106.7	%
Sustainer impulse	100	2	95	100.1	106.6	%
Booster dry mass	100	2	93.98	100	105.3	%
Sustainer dry mass	100	2	94.51	100.1	105.8	%
Coast before ignition	10	0.5	8.56	10.01	11.56	s
Sustainer fin cant	0	0.2	-0.4895	-0.009021	0.5442	deg
Booster fin cant	0	0.2	-0.6088	0.01278	0.5824	deg
Launch angle	90	0.5	88.46	89.61	90	deg
Heading	0	180	-179.6	0.3262	179.3	deg
Sustainer fin alignment	0	0.1	-0.3115	-8.15e-05	0.3489	deg, each panel
Booster fin alignment	0	0.1	-0.2804	-0.002951	0.3697	deg, each panel
Sustainer thrust offset	0	0.1	-0.3453	-0.0003712	0.325	in, each axis
Booster thrust offset	0	0.2	-0.7142	-0.008199	0.6331	in, each axis
Sustainer CG offset	0.008	0.01	-0.02861	0.0007309	0.03647	in, each axis, off the nominal
Booster CG offset	0.02611	0.02	-0.07002	0.0009234	0.06987	in, each axis, off the nominal
Sustainer product of inertia	0.9436	0.05	-0.1638	0.001983	0.1463	lb in², each axis, off the nominal
Booster product of inertia	0	0.05	-0.1556	0.002243	0.172	lb in², each axis
Forecast day	2026-09-16	one of the 15 days above, picked at random				

PANDORA - EAYF_CONFIG_1 - Inputs

```
[Case]
  Units               IN
  Analyze             STACK

[Nose]
  Shape               CONICAL
  Length              15
  PowerExponent       0.5
  TipRadius           0.1
  FlatTip             false

[Body]
  Diameter            3
  Length              51.3
  BoattailLength      0
  BoattailDiameter    3
  NozzleExitDiameter  2.5

[Fins]
  Count               4
  RootChord           8
  TipChord            5.5
  Span                3.62
  Sweep               5
  AftOffset           0
  Thickness           0.157
  Section             CHAMFER
  LEChamfer           0.75
  TEChamfer           0.75
  Cant                0

[FinCan]
  Enabled             true
  Diameter            3.15
  Length              9.16
  AftOffset           0
  FrontBevel          1.09

[Booster]
  Enabled             true
  Diameter            3
  Length              40
  Insertion           0
  TransitionLength    1
  NoseCone            false
  NoseShape           OGIVE
  NoseLength          8
  BoattailLength      0
  BoattailDiameter    0.01
  NozzleExitDiameter  2.75

[BoosterFins]
  Count               4
  RootChord           8
  TipChord            4
  Span                4.5
  Sweep               4
  AftOffset           0
  Thickness           0.118
  Section             CHAMFER
  LEChamfer           1
  TEChamfer           0.75
  Cant                0

[BoosterFinCan]
  Enabled             true
  Diameter            3.1
  Length              8.5
  AftOffset           0
  FrontBevel          0.5

[Protuberance1]
  Type                FAIRING
  Stage               SUSTAINER
  X                   17
  Count               2
  Length              0.75
  Width               0.4
  Height              0.4
  Standoff            0

[Flight]
  Altitude            4000
  TurbulentBoundaryLayerfalse
  Roughness           0.000984252

[Motors]
  Sustainer           EAYF Sustainer - N3051
  Booster             EAYF Booster - M3140.e

[Site]
  Weather             OPENMETEO
  Latitude            35.3471
  Longitude           -117.808
  Elevation           2000
  Date                2026-09-16
  Time                17:00
  WindSpeed           0
  WindDirection       0
  CylinderRadius      7

[Launch]
  RailLength          15
  Angle               90
  Heading             0
  Coast               10
  MaxTilt             15
  MinIgnitionAltitude 0
  MaxTime             1000

[Recovery]
  ApogeeChuteDiameter 8
  ApogeeChuteCd       0.8
  MainChuteDiameter   28
  MainChuteCd         1.8
  MainChuteAltitude   2000

[Thermal]
  IncludeDescent      false

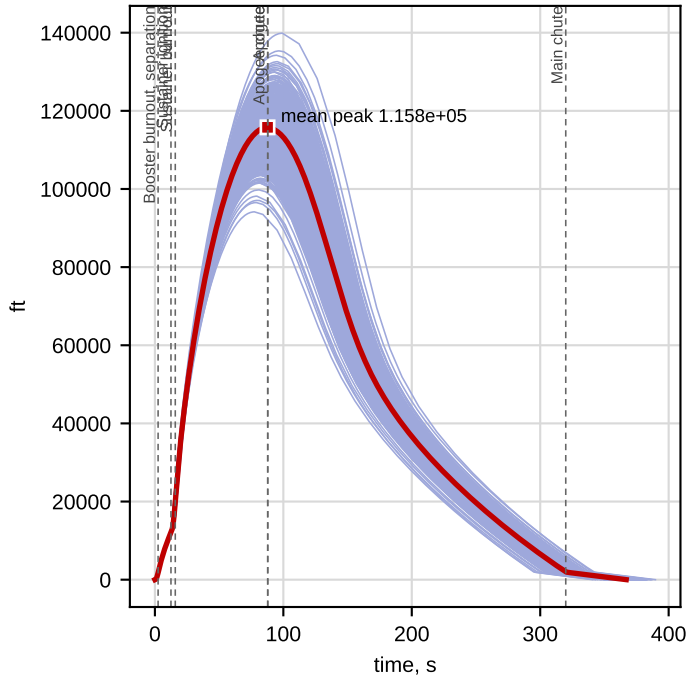
[DrySustainer]
  Mass                13.5
  CG                  42
  Inertia             4985.17
  RollInertia         37.744
  CgOffset            0.008
  ProductFin1         0.601
  Product90           0.943588
  MotorOffset         2.6

[DryStack]
  Mass                21.62
  CG                  67
  Inertia             19078
  RollInertia         63.65
  CgOffset            0.0148
  ProductFin1         1.105
  Product90           21.917
  MotorOffset         3.44

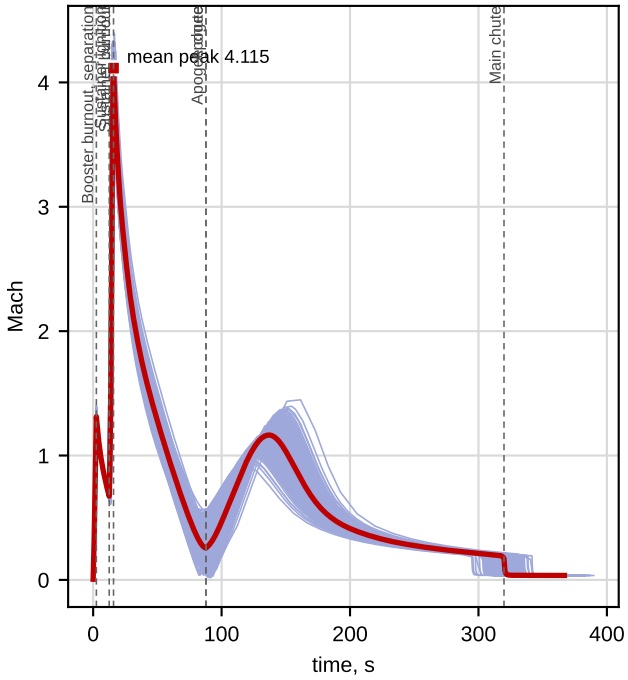
[MonteCarlo]
  Runs                512
  FailWhen            lock-in = true; resona
  ToleranceSweep      false
  AngleStd            0.5
  HeadingStd          180
  CoastStd            0.5
  BoosterImpulseStd   2
  SustainerImpulseStd 2
  BoosterMassStd      2
  SustainerMassStd    2
  CantStd             0.2
  SustainerFinAlignStd0.1
  BoosterFinAlignStd  0.1
  SustainerThrustOffsetStd0.1
  BoosterThrustOffsetStd0.2
  SustainerCgOffsetStd0.01
  BoosterCgOffsetStd  0.02
  SustainerProductStd 0.05
  BoosterProductStd   0.05
  WindSpeedStd        0
  WindDirectionStd    0
  WindDays            7
```

EAYF_CONFIG_1 - Flight, mean of 487 Monte Carlo runs

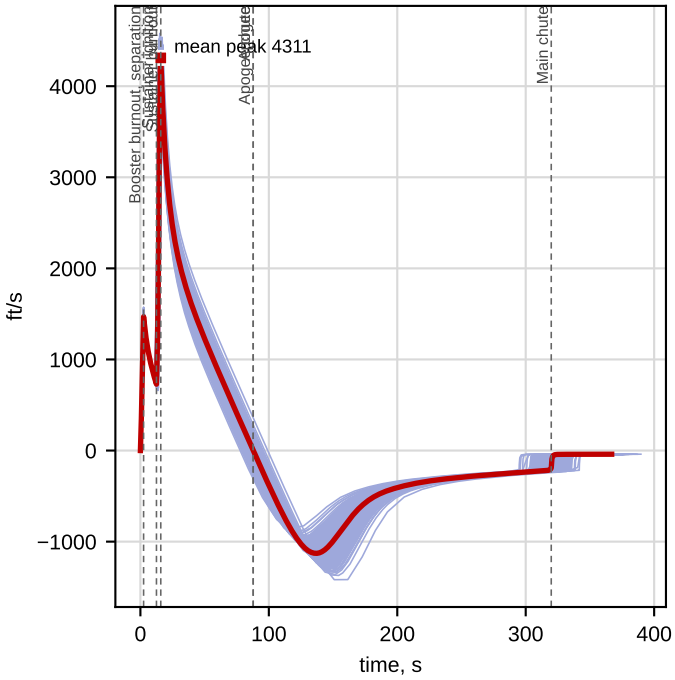
Altitude



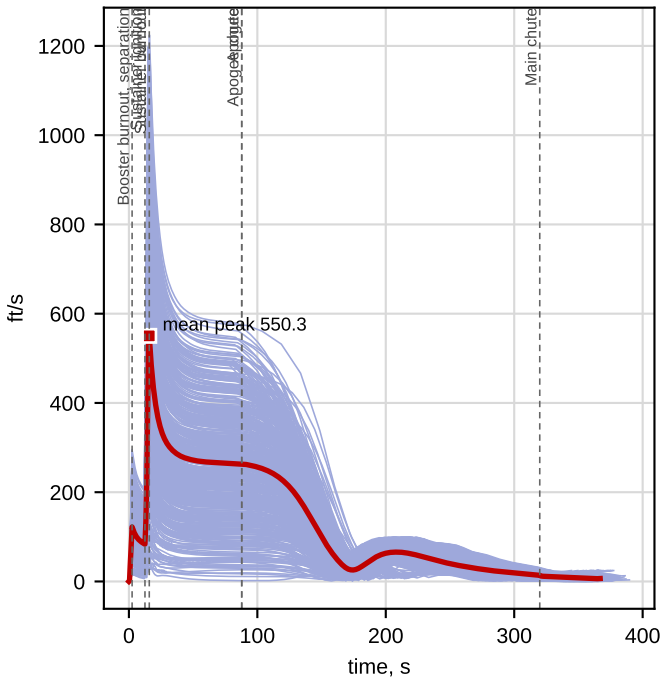
Mach number



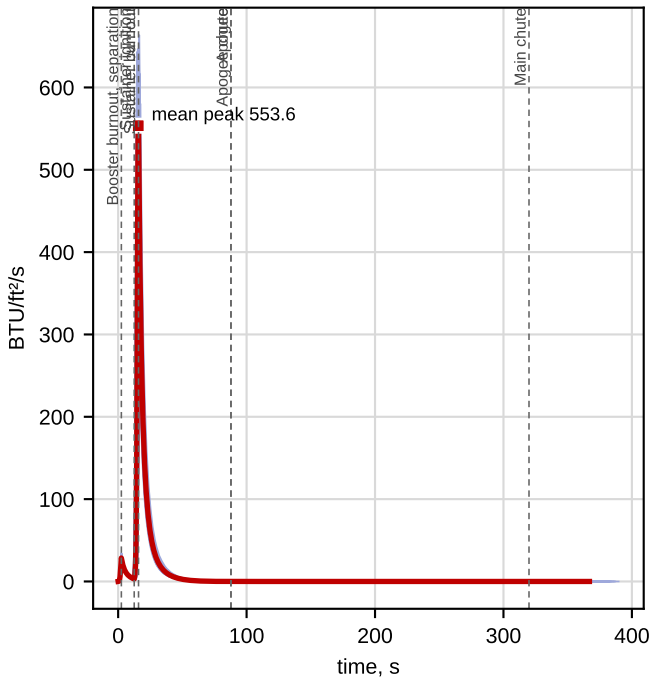
Vertical velocity



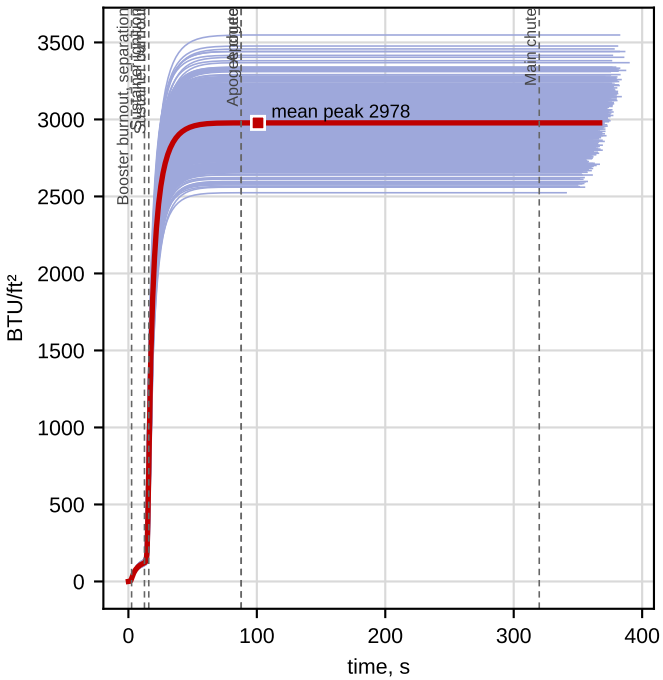
Horizontal velocity



Stagnation heat flux

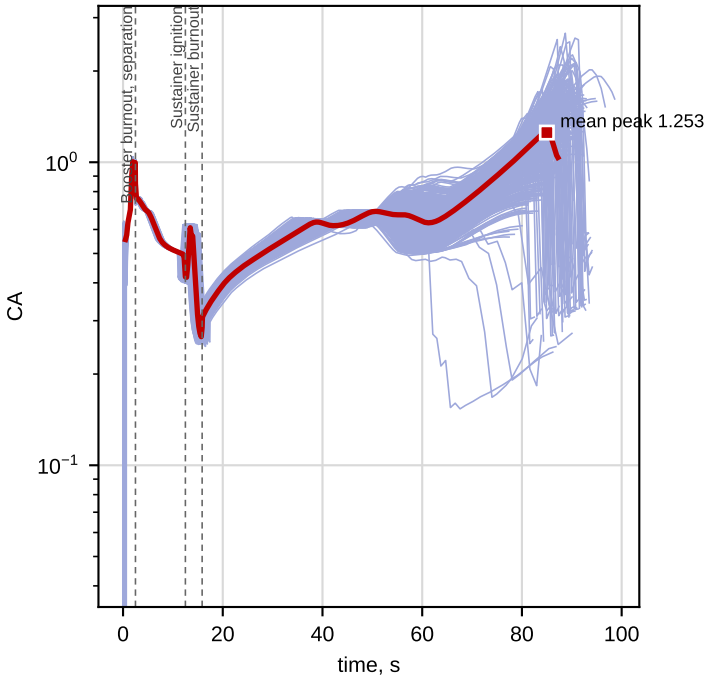


Integrated heat load

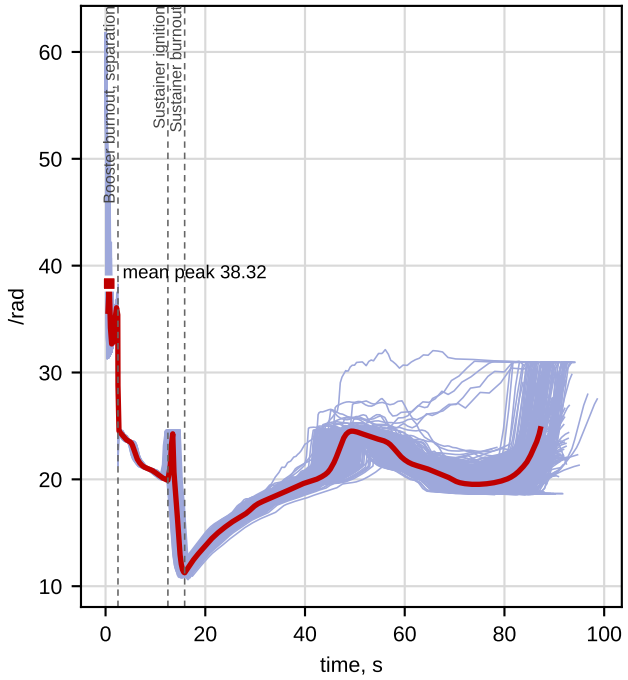


EAYF_CONFIG_1 - DATCOM aerodynamics along the flight (launch to apogee, Mach 0.1 and up)

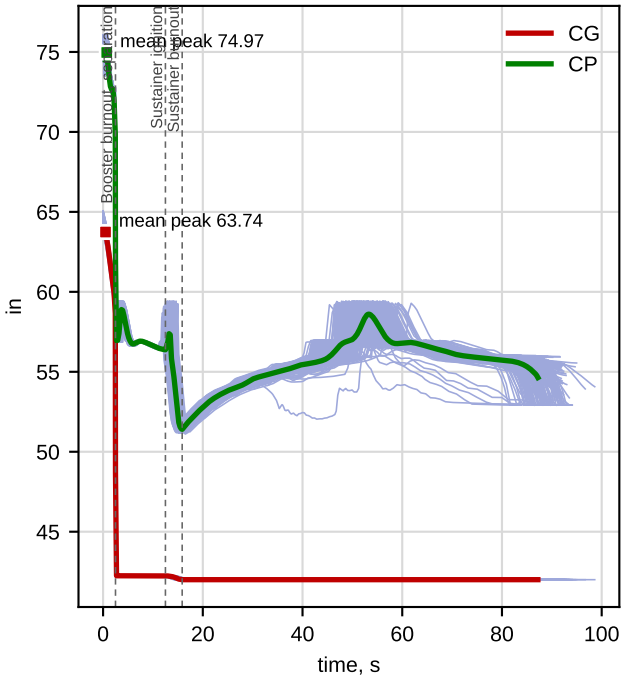
Axial force coefficient CA (log scale)



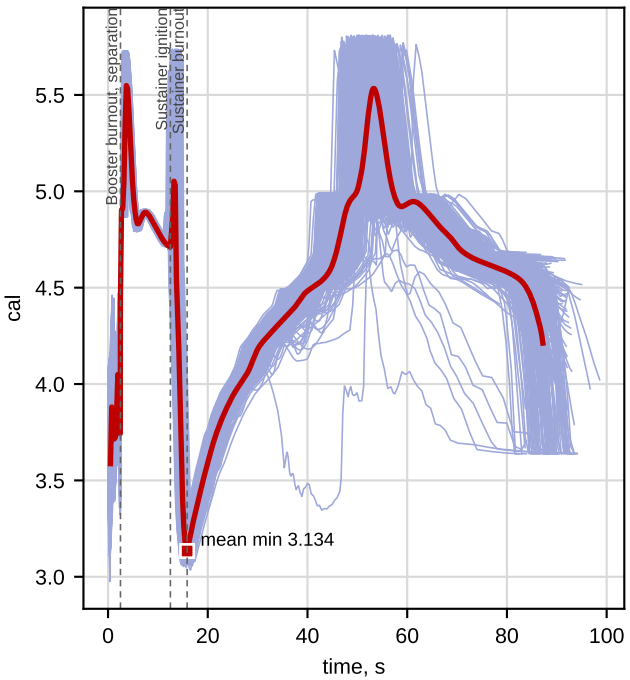
Normal force slope CNα



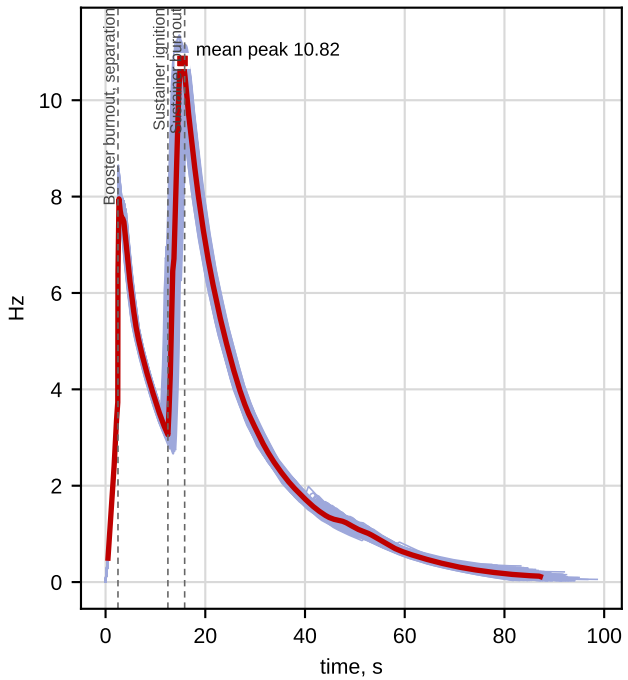
CG and centre of pressure



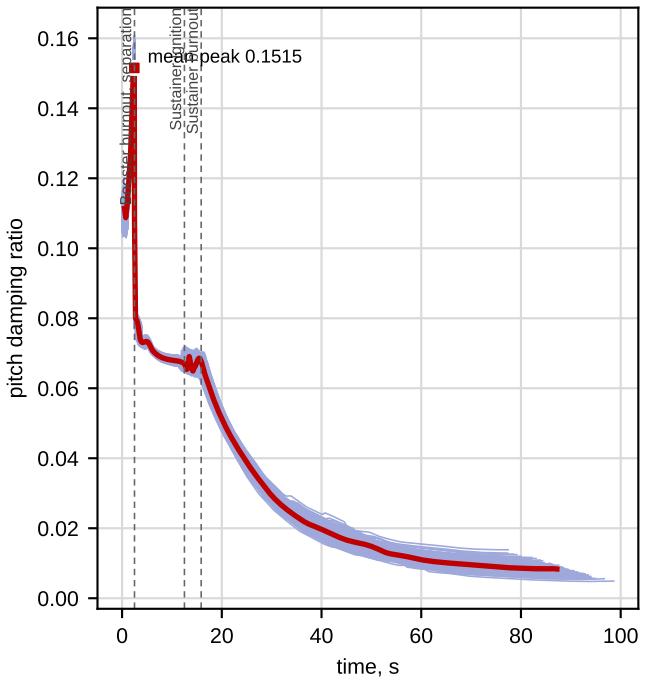
Static margin



Pitch natural frequency

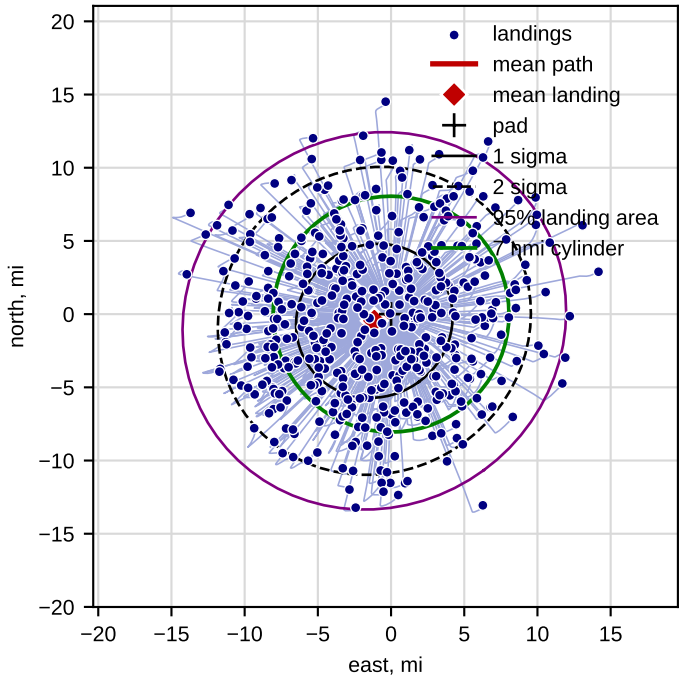


Pitch damping ratio

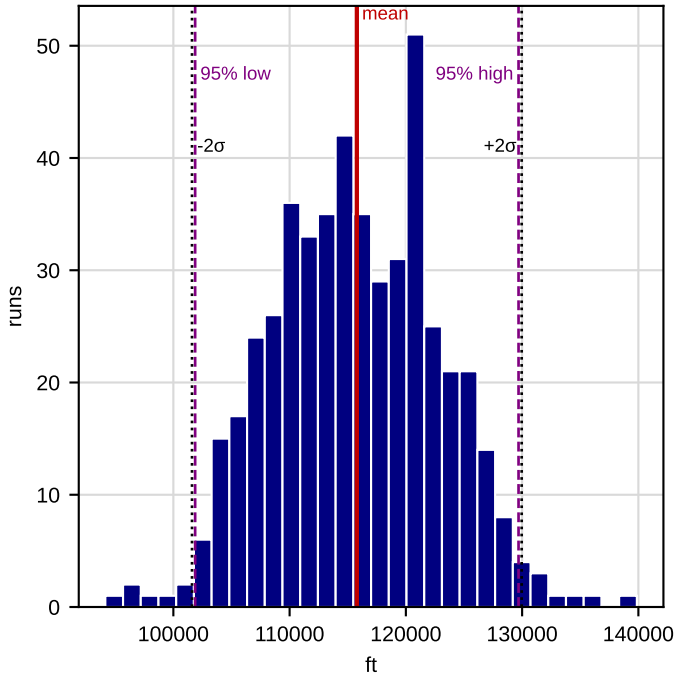


EAYF_CONFIG_1 - Monte Carlo, 487 runs (25 locked out, not shown)

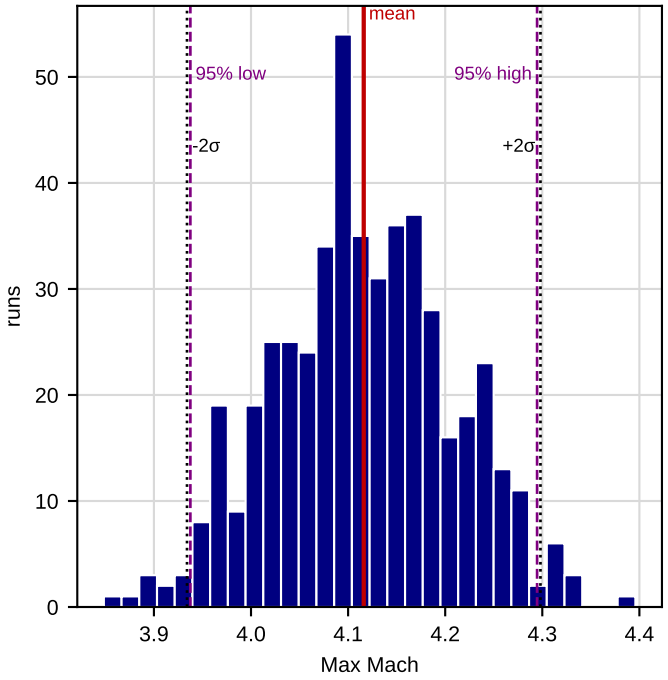
Ground tracks and landing points



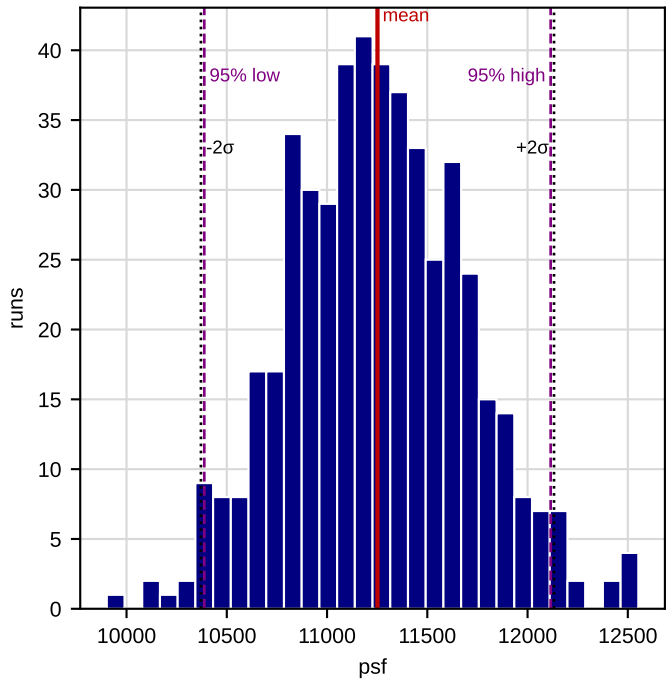
Apogee



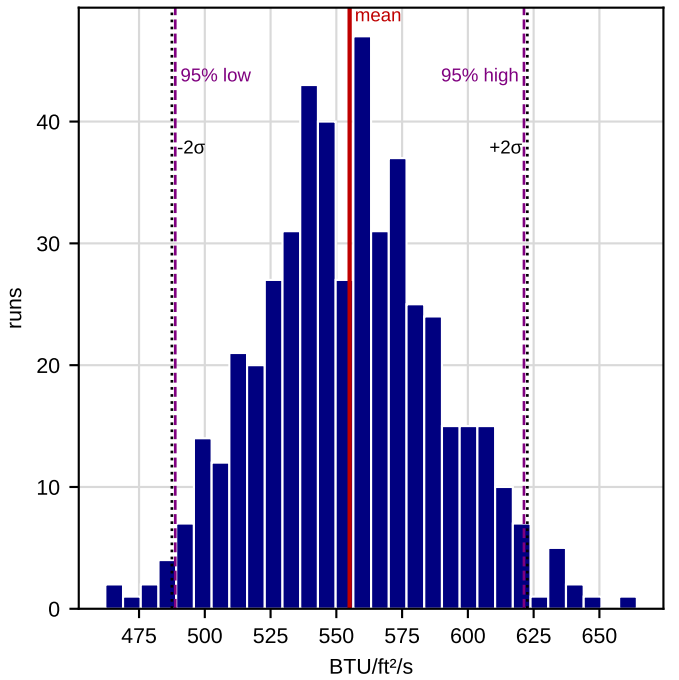
Max Mach



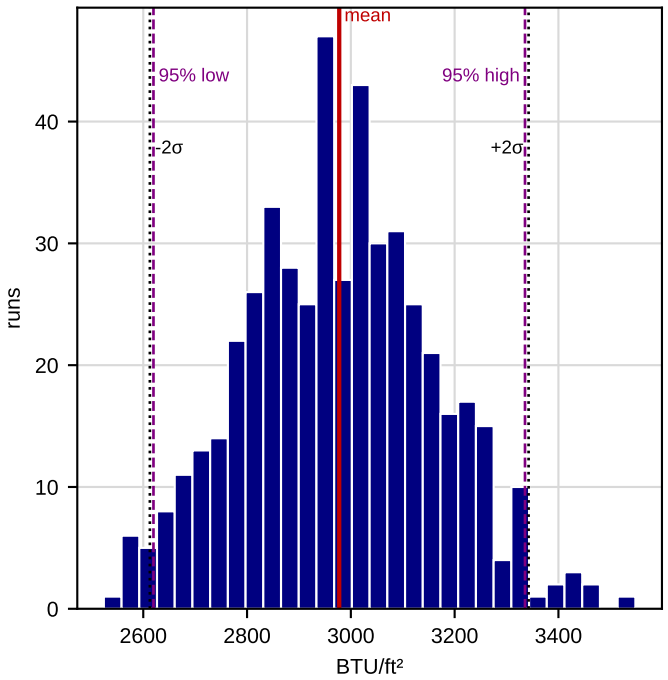
Max dynamic pressure



Peak heat flux

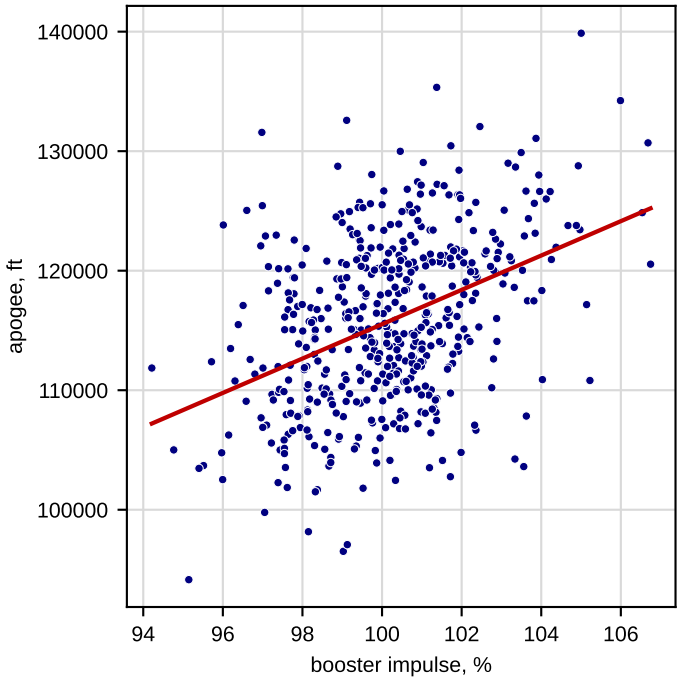


Heat load

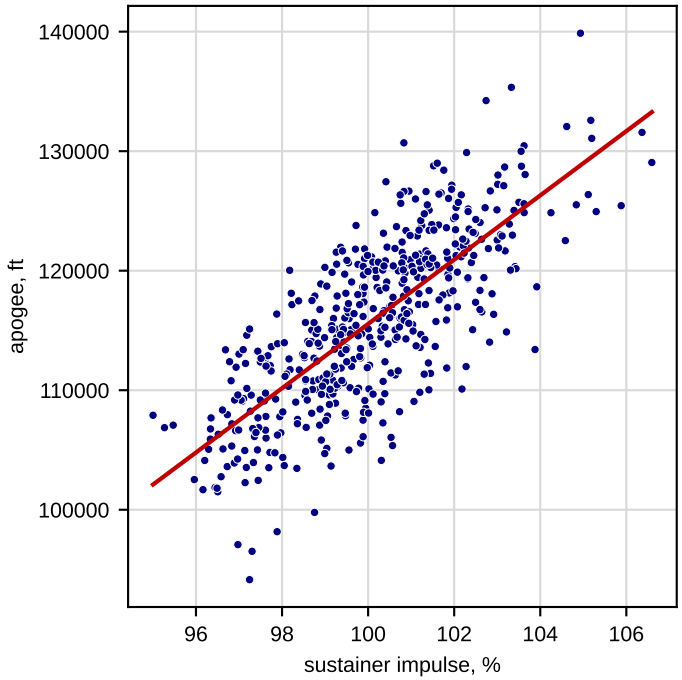


EAYF_CONFIG_1 - Sensitivity: what each varied input did to apogee, 487 runs

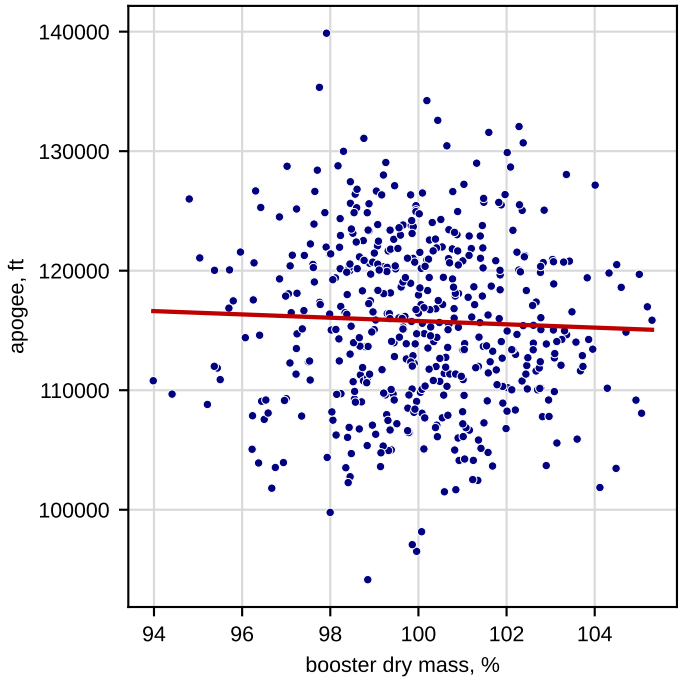
Booster impulse: +1437 ft per %



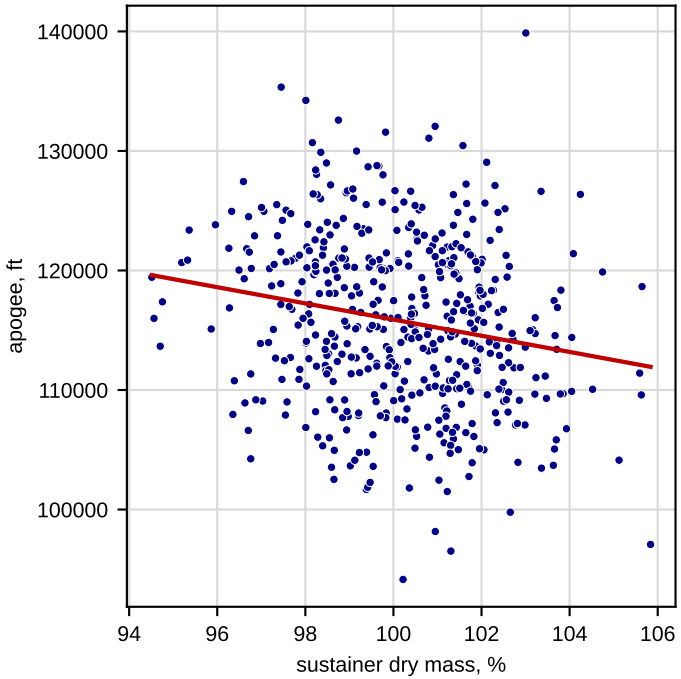
Sustainer impulse: +2689 ft per %



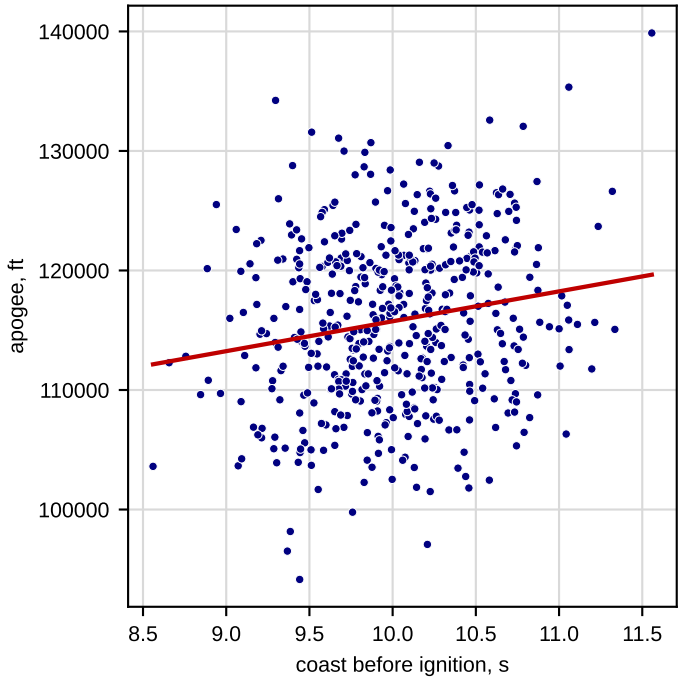
Booster dry mass: -138.1 ft per %



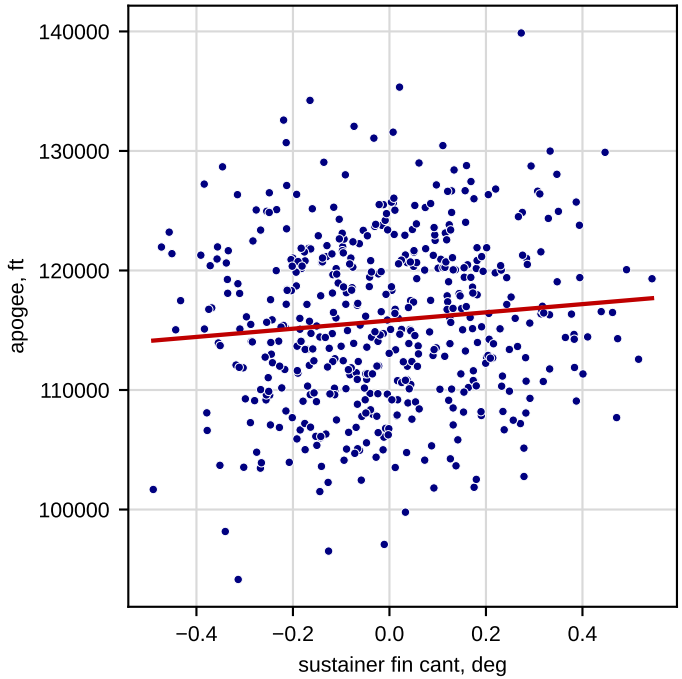
Sustainer dry mass: -677.2 ft per %

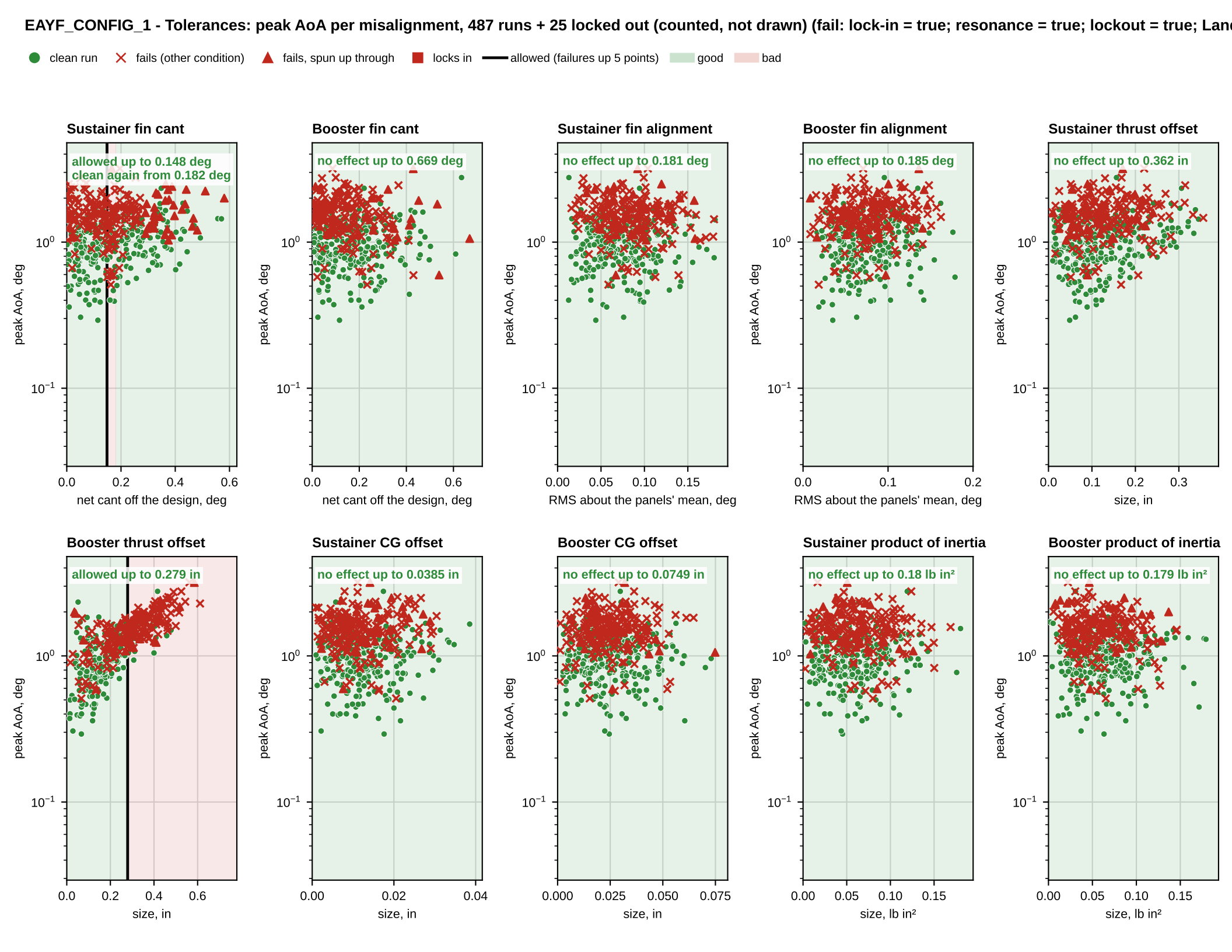


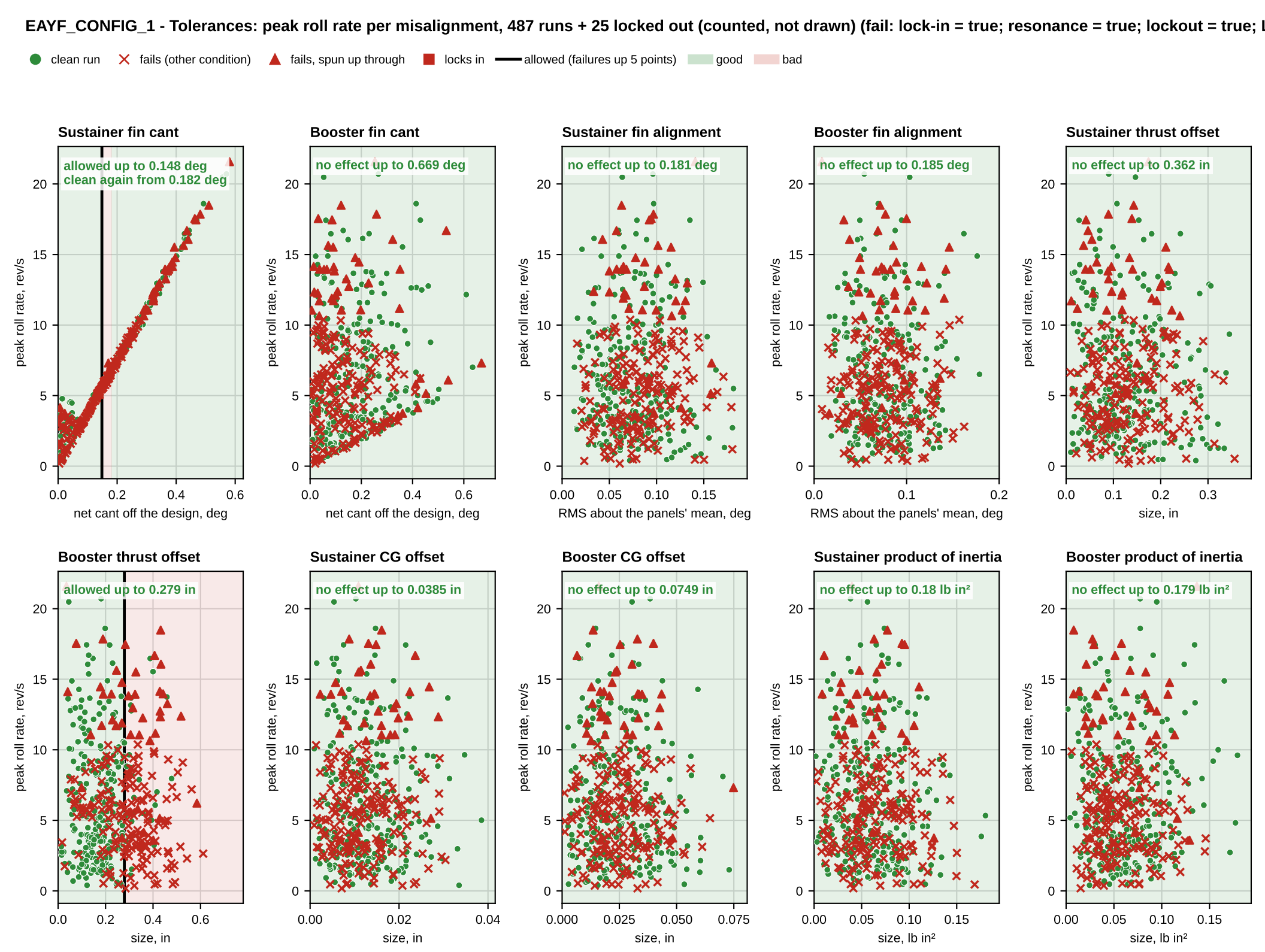
Coast before ignition: +2501 ft per s



Sustainer fin cant: +3429 ft per deg




EAYF_CONFIG_1 - Tolerances: peak AoA per misalignment, 487 runs + 25 locked out (counted, not drawn) (fail: lock-in = true; resonance = true; lockout = true; Lan
clean run
fails (other condition)
fails, spun up through
locks in
allowed (failures up 5 points)
good
bad
Sustainer fin cant
allowed up to 0.148 deg
clean again from 0.182 deg
net cant off the design, deg
Booster fin cant
no effect up to 0.669 deg
net cant off the design, deg
Sustainer fin alignment
no effect up to 0.181 deg
RMS about the panels' mean, deg
Booster fin alignment
no effect up to 0.185 deg
RMS about the panels' mean, deg
Sustainer thrust offset
no effect up to 0.362 in
size, in
Booster thrust offset
allowed up to 0.279 in
size, in
Sustainer CG offset
no effect up to 0.0385 in
size, in
Booster CG offset
no effect up to 0.0749 in
size, in
Sustainer product of inertia
no effect up to 0.18 lb in²
size, lb in²
Booster product of inertia
no effect up to 0.179 lb in²
size, lb in²
peak AoA, deg


EAYF_CONFIG_1 - Tolerances: peak roll rate per misalignment, 487 runs + 25 locked out (counted, not drawn) (fail: lock-in = true; resonance = true; lockout = true; l
clean run
fails (other condition)
fails, spun up through
locks in
allowed (failures up 5 points)
good
bad
Sustainer fin cant
allowed up to 0.148 deg
clean again from 0.182 deg
net cant off the design, deg
Booster fin cant
no effect up to 0.669 deg
net cant off the design, deg
Sustainer fin alignment
no effect up to 0.181 deg
RMS about the panels' mean, deg
Booster fin alignment
no effect up to 0.185 deg
RMS about the panels' mean, deg
Sustainer thrust offset
no effect up to 0.362 in
size, in
Booster thrust offset
allowed up to 0.279 in
size, in
Sustainer CG offset
no effect up to 0.0385 in
size, in
Booster CG offset
no effect up to 0.0749 in
size, in
Sustainer product of inertia
no effect up to 0.18 lb in²
size, lb in²
Booster product of inertia
no effect up to 0.179 lb in²
size, lb in²
peak roll rate, rev/s

EAYF_CONFIG_1 - Ground track, 487 Monte Carlo runs

